DEEPFAKE DETECTION

User Manual

Guide for Non-Technical Users

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1. What This System Does

The Deepfake Detection System lets you upload an MP4 video and instantly find out whether it is REAL (authentic footage) or FAKE (AI-generated or manipulated). The system uses a deep learning model trained to detect manipulated faces in videos.

You do not need any technical knowledge to use this system. Simply open the web application, upload your video, and read the result.

2. Getting Started

2.1 Prerequisites

- Docker and Docker Compose installed on your computer
- A modern web browser (Chrome, Firefox, Edge, or Safari)
- Host filesystem encrypted (BitLocker on Windows / LUKS on Linux)

2.2 Starting the System

Open a terminal in the deepfake-detection folder and run:

```
cp .env.example .env
# Edit .env to set your passwords (POSTGRES_PASSWORD,
GRAFANA_ADMIN_PASSWORD)
docker compose up -d
```

Wait approximately 60 seconds for all services to start. Then open your browser and go to:

Service	URL	Purpose
Main Application	http://localhost:3002	Upload videos and view results
MLflow UI	http://localhost:5000	View training experiments
Airflow UI	http://localhost:8080	View pipeline runs
Grafana	http://localhost:3001	View monitoring dashboards

3. Uploading and Analysing a Video

Step 1: Open the Application

Navigate to http://localhost:3002 in your web browser.

Step 2: Upload a Video

On the Home page, you will see a large upload area. You can either:

- Drag and drop an MP4 video file onto the upload area, OR
- Click the upload area and select an MP4 file using your file browser

Important file requirements:

Requirement	Value
File format	MP4 only
Maximum file size	100 MB
Video content	Must contain a clearly visible human face

Step 3: Wait for the Result

The system analyses your video. This typically takes 5 to 30 seconds depending on the length of the video. A loading animation will appear while the analysis is in progress.

Step 4: Read the Result

Result	Colour	Meaning
REAL	Green	The video appears to be authentic footage
FAKE	Red	The video appears to be AI-generated or manipulated

Additional information displayed with the result:

- Confidence %: How certain the model is (higher percentage = more confident)
- Grad-CAM heatmap: Shows which parts of the face influenced the decision most (red areas = high influence)
- Frames analysed: Number of video frames the model processed (up to 30)

Step 5: Submit Feedback (Optional)

After seeing the result, two buttons appear: Correct and Incorrect.

- Click Correct if the prediction matches what you know about the video
- Click Incorrect if you believe the prediction is wrong

Your feedback is saved and helps improve the model over time. A confirmation message 'Feedback recorded' will appear once submitted.

4. Pipeline Dashboard

Click Pipeline Dashboard in the navigation bar to see the system status:

Widget	Description
Pipeline Throughput	Videos processed per minute in the last pipeline run
MLflow Experiments	Recent training runs with accuracy, F1-score, and git commit
Airflow DAG Status	History of data pipeline and retraining runs

Links at the bottom of the dashboard open the full MLflow and Airflow UIs in a new tab.

5. Admin Page

Click Admin in the navigation bar to access the Admin page. This section is for technical users.

Section	Description
Current Model	The MLflow model version and run ID currently loaded for inference
Rollback	Enter a version number and click Rollback to switch to a previous model version
External Tools	Quick links to MLflow, Grafana, Prometheus, and Airflow UIs

Rolling Back to a Previous Model Version

1. Open the MLflow UI at http://localhost:5000 and note the version number you want to restore.

2. Go to the Admin page in the application.

3. Enter the version number in the Rollback to version field and click Rollback.

4. The system will reload the model — the status box will update to confirm.

6. Troubleshooting

Problem	Solution
"Only MP4 files are accepted"	Convert your video to MP4 format using a free converter
"File must be under 100MB"	Trim the video or reduce its resolution and try again
"An unexpected error occurred"	Wait 30 seconds and retry — system may still be initialising
Blank result / confidence = 0%	Video may not contain a clearly visible human face
Grafana shows no data	Wait for the first Prometheus scrape (15 second interval)
Airflow DAG not running	Check Airflow UI at localhost:8080; ensure videos are in data/landing/
Upload stuck on loading	Check that all Docker containers are running: docker compose ps

7. Stopping the System

To stop all services, run the following command in the deepfake-detection folder:

```
docker compose down
```

Your data and model files are preserved. To start again, run docker compose up -d.